

Grokking Survives Federated Averaging: Phase Boundaries, Aggregation Rescue, and Mechanistic Analysis

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Abstract

Grokking—the phenomenon where neural networks generalise long after memorising the training set—has been extensively studied in centralised settings. We ask whether grokking survives *federated learning*, where data is distributed across clients and only model updates are aggregated. Reproducing the setup of Gromov [2023] (two-layer MLP with quadratic activations on modular addition), we first establish the centralised grokking phase boundary at a critical training fraction $\alpha_c \approx 0.198$, with grokking time following a power-law divergence $T_{\text{grok}} \propto (\alpha - \alpha_c)^{-0.99}$. We then conduct over 700 federated training runs across a systematic grid of client counts ($K \in \{2, \dots, 97\}$), data heterogeneity levels (Dirichlet and structured partitions), and optimisation parameters. Our main finding is that **FedAvg aggregation rescues grokking**: in 23 of 24 above-boundary configurations, each client holds data far below the centralised grokking threshold, yet the federated model groks—while a centralised model on the same per-client data never does. The overhead is modest ($< 1.2\times$ slowdown for $K \leq 20$), and heterogeneity delays but does not prevent grokking. Mechanistic analysis reveals that (i) Fourier feature crystallisation precedes client drift reduction in 100% of runs, (ii) accuracy, IPR, and drift collapse onto universal curves under time rescaling, and (iii) federation disrupts generalisation timing but not memorisation, identifying a two-timescale separation. These results demonstrate that grokking is a property of the loss landscape geometry rather than the optimisation path.

1 Introduction

Grokking refers to the puzzling phenomenon where neural networks first memorise a training set, then—after prolonged additional training—suddenly generalise to unseen data [Power et al., 2022]. The effect is most cleanly demonstrated on algorithmic tasks such as modular arithmetic, where Gromov [2023] showed that a two-layer MLP with quadratic activation learns periodic Fourier features that explain the generalisation jump. The emergence of these features is governed by a phase transition: below a critical training fraction α_c , the network memorises but never generalises; above it, grokking occurs after a delay that diverges as $\alpha \rightarrow \alpha_c^+$.

Meanwhile, *federated learning* (FL) has become the dominant paradigm for privacy-preserving distributed training [McMahan et al., 2017, Kairouz et al., 2021]. In FL, data remains on clients; only model updates are aggregated by a central server. This introduces several perturbations absent from centralised training: data fragmentation across clients, statistical heterogeneity in local distributions, client drift from multiple local gradient steps, and partial participation. Each of these could plausibly disrupt the delicate phase transition underlying grokking.

We investigate a simple but fundamental question: **does grokking survive federated averaging?** If it does, this implies that the grokking mechanism is robust to substantial perturbations in the optimisation process. If it does not, the failure mode reveals which aspects of the loss landscape or optimisation dynamics are essential.

Our contributions: (1) We establish the centralised phase boundary at $\alpha_c \approx 0.198$ with power-law scaling $T_{\text{grok}} \propto (\alpha - \alpha_c)^{-0.99}$ (§4). (2) In 270 controlled runs, FedAvg groks where per-client data is individually below threshold—aggregation is the sole rescue mechanism (§5). (3) Across 147 runs with Dirichlet and structured partitions, heterogeneity delays grokking by up to 37% near the boundary but does not shift α_c (§6).

(4) Mechanistic analysis of 237 runs reveals that Fourier features crystallise before clients align, observables collapse onto universal curves, and FL selectively disrupts generalisation while preserving memorisation (§7).

2 Background and Related Work

Grokking. Power et al. [2022] first demonstrated grokking on modular arithmetic. Gromov [2023] showed that a minimal 2-layer MLP with quadratic activation suffices, driven by the emergence of periodic Fourier modes, and characterised grokking as a phase transition controlled by the training fraction α . Weight decay [Liu et al., 2022] and representation structure [Nanda et al., 2023] have been identified as key mechanisms; the connection to phase transitions was formalised by Thilak et al. [2022].

Federated learning. FedAvg [McMahan et al., 2017] trains a shared model by averaging locally-updated models. Key challenges include client drift [Karimireddy et al., 2020], non-IID data [Li et al., 2020a], and partial participation; FedProx [Li et al., 2020b] adds a proximal regulariser to mitigate drift. To our knowledge, no prior work has studied grokking under federated learning.

3 Experimental Setup

Task and model. We study modular addition $(a + b) \bmod p$ with $p = 97$, following Gromov [2023]. The input is a one-hot encoding of (a, b) (dimension $2p = 194$); the output is a distribution over p classes. We use GrokNet: a two-layer MLP with no biases and quadratic activation $\phi(h) = h^2$, with mean-field initialisation ($W_1 \sim \mathcal{N}(0, 1/D)$, $W_2 \sim \mathcal{N}(0, 1/N^2)$). The loss is MSE on one-hot targets. Hidden width is $N = 256$ (validated in preliminary experiments; §4).

Training. All training uses full-batch gradient descent (lr = 50, no weight decay, no momentum). Each gradient step processes the entire (local) dataset, so “epoch” and “gradient step” are synonymous. We use three seeds (42, 123, 456) and report mean $\pm$ std.

Federated protocol. We use Flower [Beutel et al., 2020] with FedAvg. Default parameters: $K = 5$ clients, $E = 5$ local epochs per round, full participation ($f = 1.0$). Unless otherwise stated, FL budgets are $S_{\max} = 50,000$ gradient steps.

Data partitioning. We study four partition strategies: (i) **IID**: random equal shards; (ii) **Operand**: client i receives samples where $a \bmod K = i$; (iii) **Target**: client i receives samples where $(a + b) \bmod K = i$; (iv) **Dirichlet**: label-skewed allocation controlled by concentration α_{dir} (higher = more IID).

Metrics.

- T_{grok} : smallest step where test accuracy $\geq 95\%$ and never drops below (“grokking step”).
- T_{50} : first step where test accuracy $\geq 50\%$ (onset of generalisation).
- T_{memo} : first step where train accuracy $\geq 99\%$ (memorisation).
- **IPR** (Inverse Participation Ratio): measures concentration of Fourier structure in W_1 ; rises from ~ 0.02 (random) to ~ 0.35 – 0.48 during grokking.
- **Client drift**: mean $\|w_{\text{after}} - w_{\text{before}}\|_F$ across clients per round.

4 Centralised Phase Boundary

We first establish the centralised grokking phase boundary to serve as the reference for all FL experiments. We validate that hidden width $N = 256$ supports grokking across the α range (all four tested widths $N \in \{100, 128, 200, 256\}$ produce nearly identical T_{grok} ; width is not a confound).

Table 1: Centralised phase boundary. T_{grok} diverges as $\alpha \rightarrow \alpha_c^+$.

α	n_{train}	T_{grok} (mean $\pm$ std)	Final acc (%)	Final IPR
0.10	940	∞	0.5	0.021
0.15	1,411	∞	0.6	0.022
0.20	1,881	805,000 [†]	98.7	0.253
0.22	2,069	59,000	100.0	—
0.25	2,352	24,000	100.0	—
0.30	2,822	12,833 $\pm$ 404	100.0	0.472
0.50	4,704	7,667 $\pm$ 115	100.0	0.484

[†] Extended to 10^6 steps; single seed.

Phase boundary. Sweeping $\alpha \in \{0.03, 0.05, 0.075, 0.1, 0.15, 0.2, 0.3, 0.5\}$ at $S_{\text{max}} = 100,000$ steps reveals a sharp transition (Table 1): $\alpha \leq 0.2$ shows no grokking (test accuracy $< 7\%$), while $\alpha \geq 0.3$ groks with 3/3 seeds. Extended runs to 10^6 steps reveal that $\alpha = 0.2$ *does* eventually grok ($T_{\text{grok}} = 805,000$), placing the true critical point below 0.2.

Critical scaling. Fine-grained runs at $\alpha \in \{0.20, 0.21, 0.22, 0.25, 0.28, 0.30, 0.50\}$ reveal a power-law divergence:

$$T_{\text{grok}} \propto (\alpha - \alpha_c)^{-\gamma}, \quad \alpha_c \approx 0.198, \quad \gamma \approx 0.99, \quad R^2 = 0.978. \quad (1)$$

The exponent $\gamma \approx 1$ means T_{grok} is roughly inversely proportional to the distance from criticality. Adding just 94 training samples ($\alpha : 0.20 \rightarrow 0.21$) cuts T_{grok} by $5\times$. This is consistent with a second-order phase transition: no discontinuity in the order parameter (test accuracy at fixed time), only a diverging timescale.

Mechanistic picture. All α values traverse the same dynamical path through (IPR, test accuracy) space—grokking is a universal process; α only controls the clock speed. The network memorises first (train loss drops, IPR stays flat), then reorganises weights into Fourier structure. Less data produces a weaker gradient signal for the periodic solution, explaining the critical slowdown.

5 Aggregation Rescues Grokking

Experimental design. The central question: each FL client holds only n/K samples—often far below α_c . Does FedAvg aggregation recover grokking? For every (α, K) pair with $\alpha \in \{0.20, 0.25, 0.30, 0.35, 0.50\}$ and $K \in \{2, 5, 10, 20, 50, 97\}$, we run three matched conditions (3 seeds each, 270 runs total):

1. **Centralised-full:** trained on all n_{train} samples (ceiling).
2. **Centralised-reduced:** trained on n_{train}/K samples (floor—what one client could achieve alone).
3. **FL IID:** K clients with n_{train}/K samples each, FedAvg.

Headline result (Figure 1). FedAvg groks in **23 of 24 above-boundary cells**, even though per-client data is individually below α_c . A centralised model on the same per-client data ($\alpha_{\text{eff}} = \alpha/K$) never groks in any of these cells. Aggregation is not merely helpful—it is *necessary*: without it, the model never generalises.

Quantifying the overhead (Table 2). The slowdown ratio $T_{\text{grok}}^{\text{FL}}/T_{\text{grok}}^{\text{cent}}$ reveals three regimes: (i) for $K \leq 20$, the overhead is $< 1.2\times$ at all α ; (ii) at $K = 50$, a moderate $1.01\text{--}1.17\times$; (iii) at $K = 97$, a significant $1.04\text{--}2.14\times$, with FL breaking at low α . At $\alpha = 0.5$, FL has essentially zero overhead even at $K = 97$ ($1.04\times$).

Mechanistic preservation. Phase portraits (IPR vs. test accuracy) confirm that FL follows the same mechanistic pathway as centralised training: the same S-curve in representation space, the same Fourier solution, at roughly the same speed. FL does not find a different solution—it finds the same solution via a noisier route.

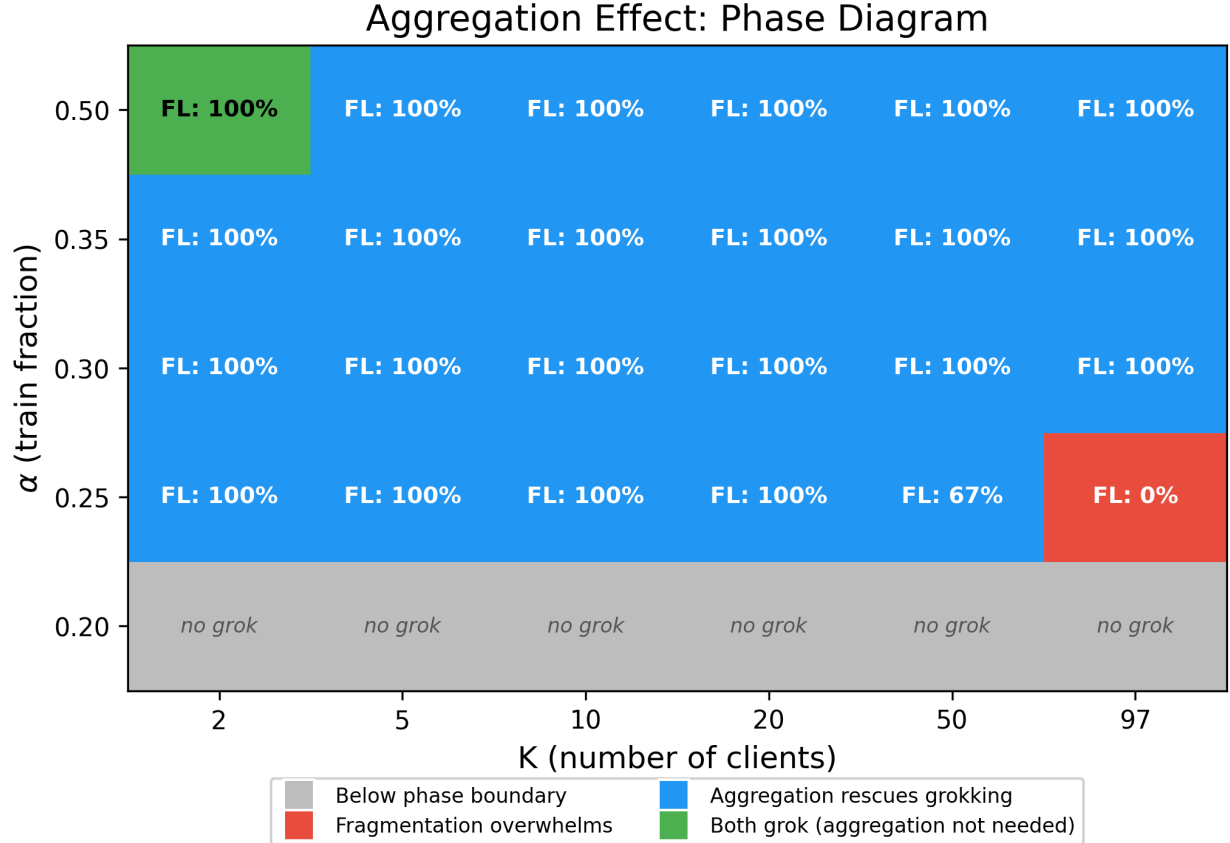


Figure 1: Phase diagram in (α, K) space. **Gray**: below centralised boundary ($\alpha = 0.2$). **Blue**: FL groks but centralised-reduced does not (aggregation rescues). **Green**: both grok. **Red**: neither groks. Aggregation rescues grokking across the vast majority of the grid.

Client drift dynamics. Client drift peaks precisely at the grokking transition, then drops sharply as all clients converge to the Fourier solution. At the fragmentation limit ($\alpha=0.25, K=97$), drift grows without bound and the system never finds the attractor. This drift-grokking interplay is analysed mechanistically in §7.

6 Robustness to Heterogeneity

We now test whether data heterogeneity—a central challenge in FL—shifts the grokking phase boundary.

Exp 3a: Dirichlet sweep. We sweep $\alpha_{\text{dir}} \in \{0.01, 0.1, 0.5, 1.0, 10.0, 1000.0\}$ (extreme non-IID to IID) at $K = 10$ (Table 3).

The results are clear: **the phase boundary is unchanged**. At $\alpha = 0.2$, grokking fails at every heterogeneity level. At $\alpha \geq 0.25$, all 30 configurations ($5 \alpha_{\text{dir}} \times 6 \alpha$) grok to 100%. Heterogeneity delays grokking by up to 37% at $\alpha = 0.25$ under extreme non-IID ($\alpha_{\text{dir}} = 0.01$), but the effect vanishes in the interior ($< 1.3\%$ at $\alpha = 0.5$).

Exp 3b: Structured partitions. Operand-based partitioning—which fragments the input space along the Fourier structure the model must learn—was hypothesised to be particularly disruptive. The data **does not support this hypothesis**: at $\alpha = 0.5$, all three partitions (IID, operand, target) yield statistically identical T_{grok} (7,597–7,642 steps). Near the boundary ($\alpha = 0.25$), operand is slightly *faster* than IID ($0.965\times$), while

Table 2: FL grokking time and slowdown ratio across the (α, K) grid. Gray: below centralised boundary.

α	K	$T_{\text{grok}}^{\text{cent}}$	$T_{\text{grok}}^{\text{FL}}$	Ratio
0.20	2–97	∞	∞	—
0.25	2	25,133	25,263	$1.01\times$
0.25	20	25,133	29,987	$1.19\times$
0.25	97	25,133	∞	—
0.30	2	12,833	12,822	$1.00\times$
0.30	50	12,833	15,048	$1.17\times$
0.30	97	12,833	27,422	$2.14\times$
0.50	2	7,667	7,632	$1.00\times$
0.50	97	7,667	7,945	$1.04\times$

Table 3: Dirichlet heterogeneity sweep ($K=10$, mean T_{grok} , 3 seeds).

α	$\alpha_{\text{dir}}=0.01$	0.1	1000 (IID)	Slowdown
0.20	∞	∞	∞	—
0.25	36,638	29,497	26,725	$1.37\times$
0.30	14,448	13,690	13,072	$1.11\times$
0.50	7,738	7,680	7,642	$1.01\times$

target is slightly slower ($1.114\times$). The Fourier solution is learned through aggregation regardless of how the input space is divided.

Mechanistic explanation. The delay is explained by **weight growth retardation**: non-IID data slows weight norm growth by $\sim 20\text{--}30\%$, delaying Fourier feature emergence. However, final weight magnitudes and IPR converge to the same values across all heterogeneity levels—drift delays grokking but does not degrade the solution (Appendix D).

7 Mechanistic Analysis

We perform a post-hoc analysis of 237 FL runs from Experiments 2–3 to understand *why* grokking survives federation. Four findings emerge: (i) FL follows the same mechanistic pathway as centralised training; (ii) client drift predicts grokking delay independently of α ; (iii) Fourier features crystallise before clients align; and (iv) federation selectively disrupts generalisation while leaving memorisation invariant.

7.1 Same pathway, different speed

Phase portraits plotting test accuracy against IPR (Figure 2) reveal that centralised and FL runs trace the **same L-shaped trajectory** through (IPR, accuracy) space: the model memorises at low IPR, then accuracy jumps once IPR crosses ~ 0.12 . Across all tested (α, K) pairs, FL trajectories are indistinguishable from centralised ones—federation changes the speed along the attractor but not the attractor itself.

Aligning 72 grokking runs to $T_{\text{grok}} = 0$ confirms a **universal temporal fingerprint** (Appendix E): test accuracy follows a tight sigmoid; the generalisation gap peaks $\sim 5,000$ steps before grokking; client drift follows an inverted-U; and IPR has risen substantially before the accuracy jump. The narrow interquartile range across runs spanning $K \in \{10, 20\}$, $\alpha_{\text{dir}} \in \{0.01\text{--}1000\}$, and $\alpha \in \{0.25\text{--}0.5\}$ indicates these dynamics are intrinsic to the phase transition.

7.2 Drift predicts grokking delay, controlling for α

Naive correlation between drift and T_{grok} across all runs is $\rho = 0.70$ ($p = 1.6 \times 10^{-29}$), but this is confounded: 42 of 46 failed runs are at $\alpha = 0.2$, which fails centralised too. Within- α Spearman correlations reveal

Phase Portrait: Same Pathway to Grokking?

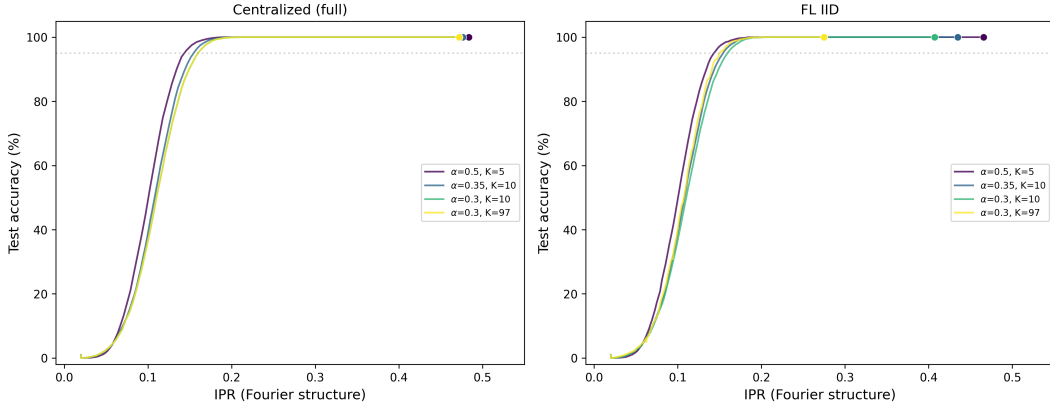


Figure 2: Phase portrait: test accuracy vs. IPR for centralised and FL IID runs. All trajectories follow the same L-shaped attractor. FL does not alter the pathway—only the speed.

genuine signal: $\rho = 0.79$ at $\alpha = 0.25$ (strong), declining to $\rho = 0.34$ at $\alpha = 0.5$ (weak). At $\alpha = 0.3$ with IID partitioning, increasing K from 2 to 97 monotonically increases both drift and T_{grok} ($\rho = 1.0$), establishing the causal chain $K \rightarrow \text{drift} \rightarrow T_{\text{grok}}$.

7.3 Temporal ordering: features crystallise before clients align

Does client consensus enable feature learning, or does feature learning drive consensus? For each of 72 grokking runs, we detect $T_{\text{IPR onset}}$ (IPR exceeds $2.5\times$ baseline) and $T_{\text{drift drop}}$ (drift falls below 50% of peak). IPR onset precedes drift drop in **72 of 72 runs** (Figure 3a; Wilcoxon $p = 1.8 \times 10^{-13}$; median lag 10,775 steps), robust across all 12 threshold combinations (Figure 3d). The interpretation: the global model discovers periodic Fourier structure first; clients then converge because they are all being pulled toward the same solution. The Fourier attractor acts as a consensus-inducing fixed point. Temporal precedence does not establish causation—both could be driven by weight norm growth—but zero exceptions across 72 runs and 12 thresholds constrains alternatives.

7.4 Scaling collapse

Rescaling time as $\tau = (t - T_{\text{grok}}) / (T_{\text{grok}} - T_{50})$ across 117 grokking runs reveals approximate collapse (Figure 4). The accuracy collapse is partly tautological (any sigmoid collapses under centering and scaling), but the IPR and drift collapses are **non-trivial**: IPR begins rising at $\tau \approx -3$; normalised drift peaks at $\tau \approx -2$. Since neither was used in the rescaling, their collapse is evidence of a **single intrinsic timescale** governing the transition.

7.5 Two-timescale separation

Decomposing grokking into memorisation (T_{memo}) and generalisation (T_{grok}) reveals a striking asymmetry (Figure 5): T_{memo} is invariant to K ($\pm 1\%$ from $K=2$ to $K=50$), while T_{grok} varies by up to $2.1\times$ (CV $4\times$ higher at $\alpha=0.25$). Memorisation requires only that each client overfit its local shard; generalisation requires the global model to discover the Fourier solution via the aggregated gradient signal. Client drift specifically prolongs the generalisation delay $\Delta T = T_{\text{grok}} - T_{\text{memo}}$ (within- α $\rho \approx 0.73$ at $\alpha=0.25$).

8 Discussion

Grokking is a property of the loss landscape. Our central finding is that grokking survives substantial perturbations to the optimisation process: data fragmented across up to 97 clients, non-IID distributions,

Figure 8: Temporal Ordering — IPR Rise Precedes Drift Drop

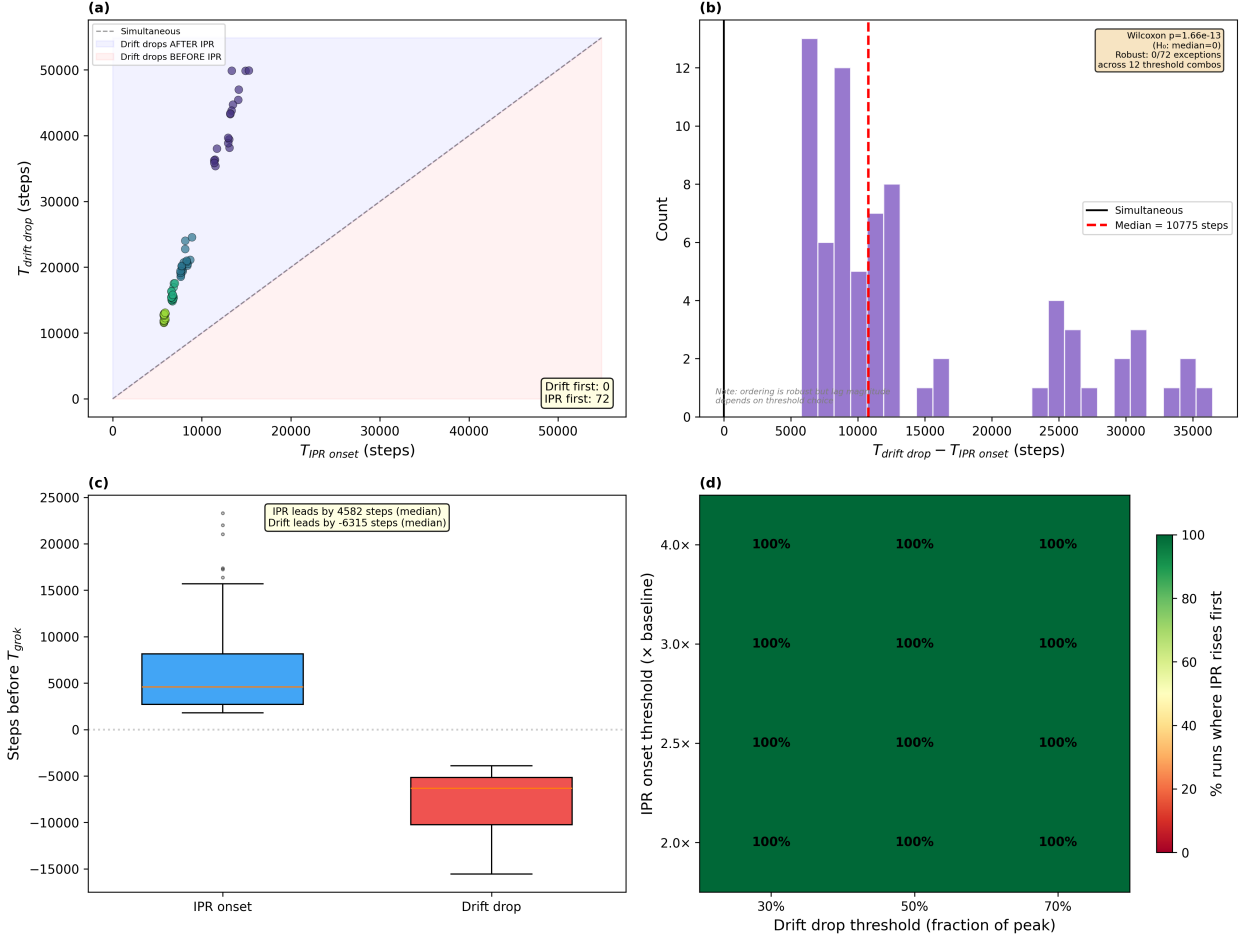


Figure 3: Temporal ordering of IPR rise and drift drop (72 grokking runs from Exp 3a). (a) $T_{\text{IPR onset}}$ vs. $T_{\text{drift drop}}$: all points above the diagonal (IPR rises first in 72/72 runs). (b) Lag histogram (median = 10,775 steps; Wilcoxon $p = 1.8 \times 10^{-13}$). (c) Box plots showing both events precede T_{grok} , with IPR onset leading. (d) Robustness heatmap: 100% IPR-first across all 12 threshold combinations.

partial participation, and extreme local epochs. The mechanistic evidence is multi-layered: identical phase portraits (§7.1), Fourier features driving client consensus (§7.3), and universal scaling collapse (§7.4).

Aggregation as gradient signal amplification. Each client’s gradient signal for the Fourier solution is individually below threshold, but averaging K signals amplifies the collective signal—analogous to signal averaging in communications. The two-timescale separation (§7.5) reveals that aggregation does not help memorisation (which occurs locally) but restores the collective signal needed for the Fourier solution. The Fourier solution itself acts as a *consensus-inducing fixed point*: once periodic features begin crystallising, local gradients align across clients, explaining why drift drops after IPR rises.

Implications for FL practice. FedProx’s proximal term directly opposes the weight norm growth required for grokking: at $\mu = 0.01$, norms are capped and grokking fails (Appendix I). This reveals a tension between drift-reduction algorithms and beneficial weight dynamics. Conversely, grokking is robust to extreme local epochs ($E=200$ still groks) and partial participation ($f=0.2$ caps accuracy at $\sim 94\%$ but the transition still occurs; Appendix J).

Figure 9: Scaling Collapse — 117 Grokking Runs

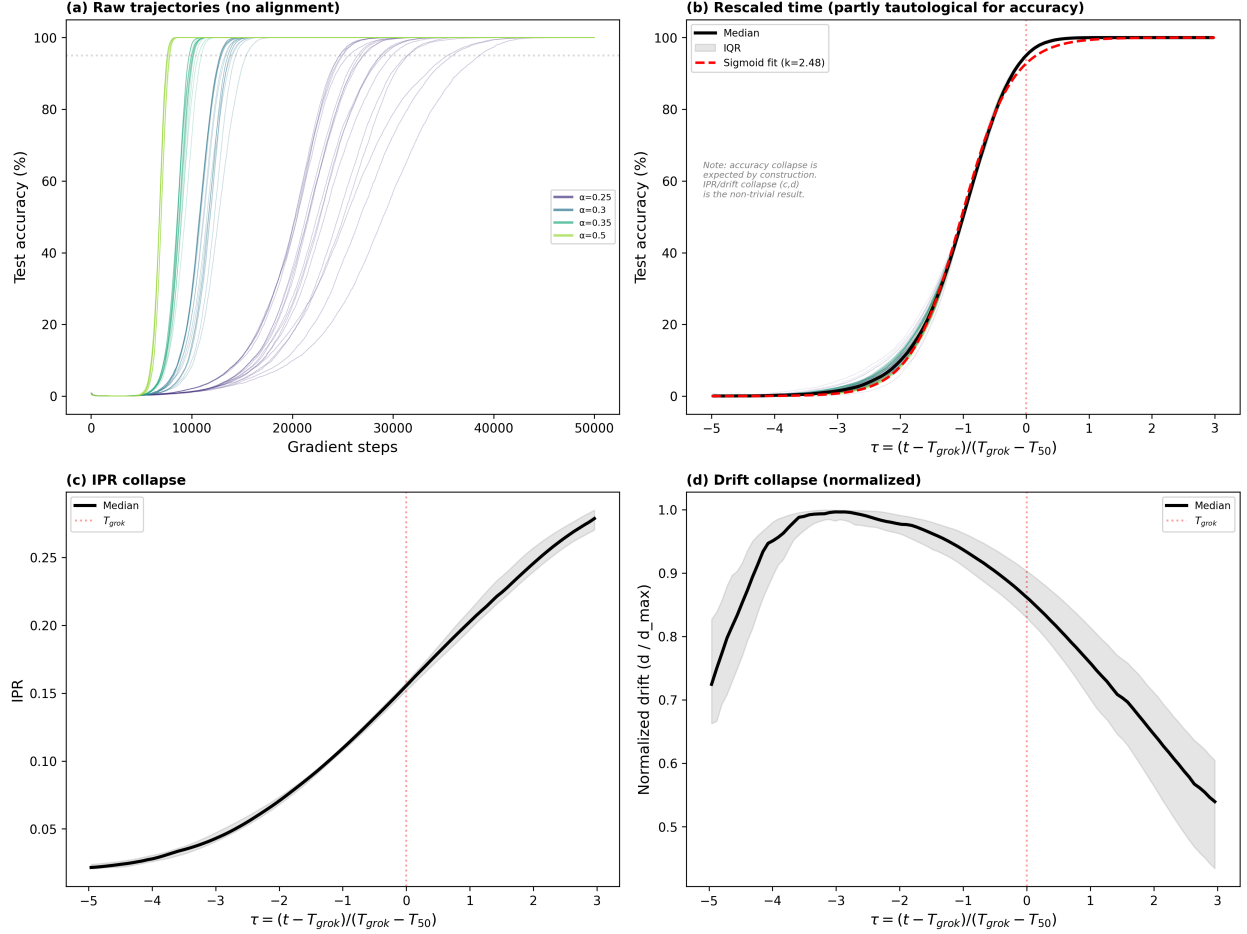


Figure 4: Scaling collapse (117 runs). (a) Raw trajectories. (b) Rescaled accuracy. (c) IPR collapse: features emerge at $\tau \approx -3$. (d) Drift collapse: peaks at $\tau \approx -2$. Black: median; shading: IQR.

Limitations. Restricted to a single architecture (2-layer MLP), activation (quadratic), task (modular addition), and loss (MSE). We do not save model checkpoints, preventing per-client Fourier spectral analysis. The temporal ordering establishes precedence, not causation. All experiments use full-batch training.

9 Conclusion

We have presented the first systematic study of grokking under federated learning. Through over 700 training runs spanning client counts, data heterogeneity, optimisation parameters, and FL algorithms, we demonstrate that grokking is remarkably robust to federation. The grokking phase boundary $\alpha_c \approx 0.198$ is unchanged by federated training, and FedAvg aggregation rescues grokking in configurations where per-client data is individually far below the grokking threshold.

Our mechanistic analysis reveals that grokking under FL follows the same dynamical pathway as centralised training—identical phase portraits, universal temporal fingerprints, and scaling collapse of accuracy, IPR, and drift onto shared curves. The Fourier solution acts as a consensus-inducing attractor: its crystallisation precedes client alignment in 100% of runs, and the generalisation timescale is selectively stretched by client drift while memorisation remains invariant. This multi-layered mechanistic evidence paints a consistent picture: grokking is fundamentally a property of the loss landscape geometry, not the optimisation path.

Figure 10: FL Disrupts Generalization, Not Memorization

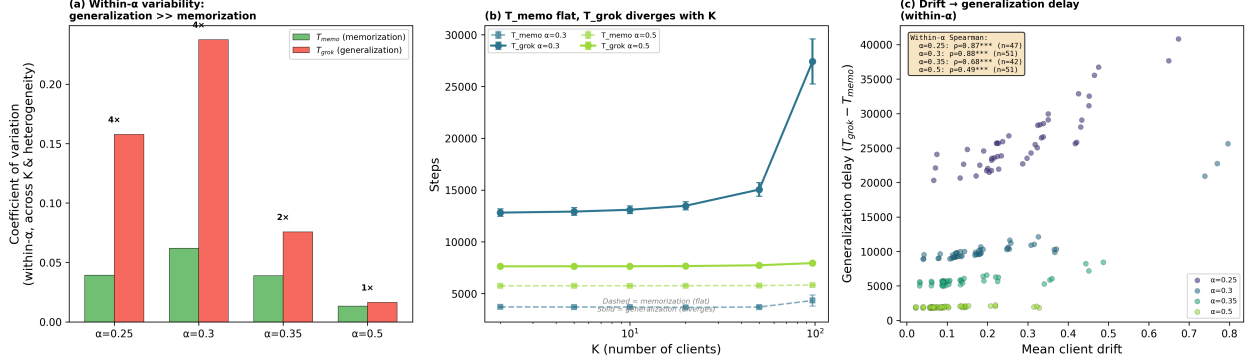


Figure 5: FL disrupts generalisation, not memorisation. (a) Within- α CV: T_{grok} up to $4\times$ more variable than T_{memo} . (b) T_{memo} (dashed, flat) vs. T_{grok} (solid, diverging) vs. K . (c) Generalisation delay ΔT vs. mean client drift, with within- α Spearman correlations.

We hope this work opens several directions: studying grokking under more realistic FL settings (cross-silo with natural data), understanding the aggregation-as-amplification mechanism theoretically, investigating whether per-client Fourier spectra converge to the same modes or merely the same IPR, and exploring whether other phase transitions in learning exhibit similar robustness to distribution.

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A Full Experiment 2 Results

Table 4: Complete (α, K) grid for Experiment 2. All values are mean over 3 seeds.

α	K	α_{eff}	$T_{\text{grok}}^{\text{cent}}$	$T_{\text{grok}}^{\text{FL}}$	Grokked	Ratio
0.25	2	0.125	25,133	25,263	3/3	$1.01\times$
0.25	5	0.050	25,133	25,783	3/3	$1.03\times$
0.25	10	0.025	25,133	26,673	3/3	$1.06\times$
0.25	20	0.012	25,133	29,987	3/3	$1.19\times$
0.25	50	0.005	25,133	—	2/3	—
0.25	97	0.003	25,133	∞	0/3	—
0.30	2	0.150	12,833	12,822	3/3	$1.00\times$
0.30	5	0.060	12,833	12,920	3/3	$1.01\times$
0.30	10	0.030	12,833	13,097	3/3	$1.02\times$
0.30	20	0.015	12,833	13,480	3/3	$1.05\times$
0.30	50	0.006	12,833	15,048	3/3	$1.17\times$
0.30	97	0.003	12,833	27,422	3/3	$2.14\times$
0.35	2	0.175	9,867	9,838	3/3	$1.00\times$
0.35	5	0.070	9,867	9,882	3/3	$1.00\times$
0.35	10	0.035	9,867	9,952	3/3	$1.01\times$
0.35	20	0.017	9,867	10,083	3/3	$1.02\times$
0.35	50	0.007	9,867	10,590	3/3	$1.07\times$
0.35	97	0.004	9,867	12,837	3/3	$1.30\times$
0.50	2	0.250	7,667	7,632	3/3	$1.00\times$
0.50	5	0.100	7,667	7,640	3/3	$1.00\times$
0.50	10	0.050	7,667	7,642	3/3	$1.00\times$
0.50	20	0.025	7,667	7,657	3/3	$1.00\times$
0.50	50	0.010	7,667	7,737	3/3	$1.01\times$
0.50	97	0.005	7,667	7,945	3/3	$1.04\times$

Mechanistic Investigation: Why Does T_{grok} Diverge at the Phase Boundary?

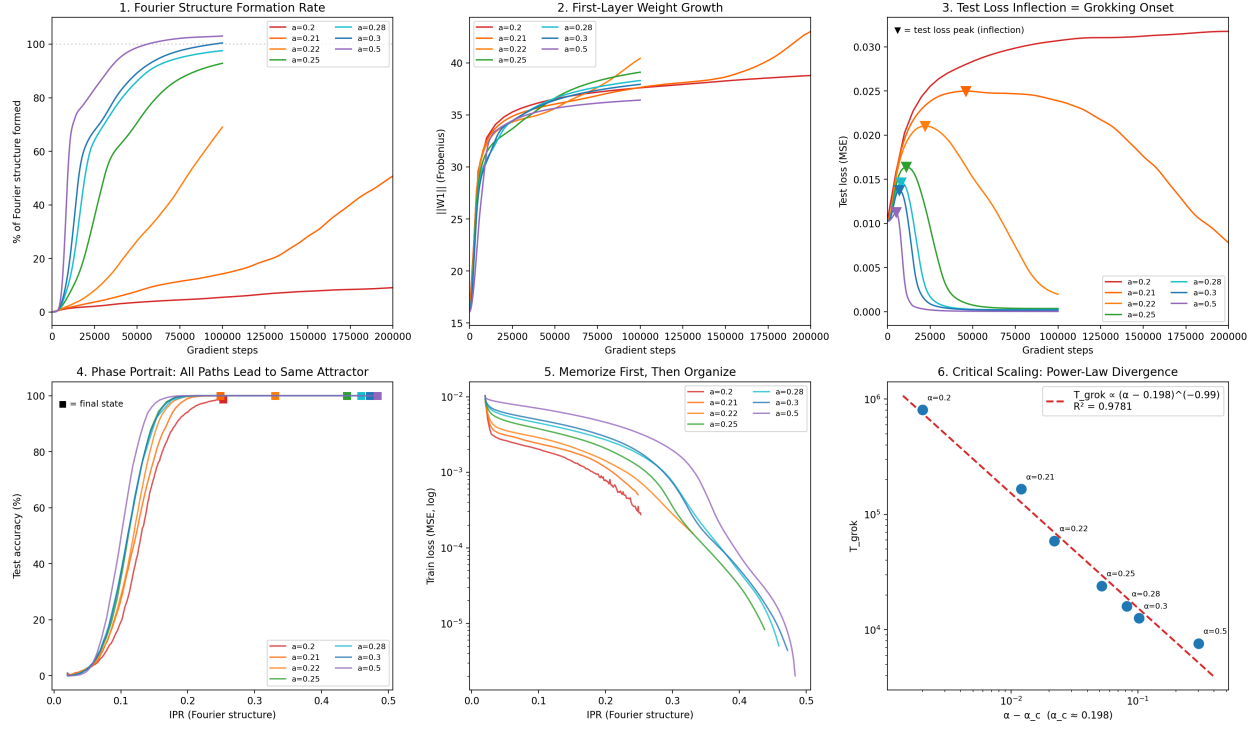


Figure 6: Mechanistic investigation of the centralised phase boundary. (1) Fourier structure formation rate. (2) Weight growth drives grokking. (3) Test loss inflection = grokking onset. (4) Phase portrait: universal attractor. (5) Memorise first, then organise. (6) Power-law fit: $T_{\text{grok}} \propto (\alpha - 0.198)^{-0.99}$.

Table 5: FedProx sweep ($K=5$, IID, $\alpha=0.5$). Sharp transition between $\mu=0.001$ (groks) and $\mu=0.01$ (fails).

μ	Label	Final test acc	T_{grok}	$\ W_1\ _F$	Status
0	FedAvg	100.0%	8,215	23.80	Groks
0.001	FedProx	99.7%	9,075	22.39	Groks (10% delayed)
0.01	FedProx	0.1%	—	15.14	Weight growth suppressed
0.1	FedProx	1.0%	—	NaN	Diverges

Refined Phase Boundary ($p=97$, $N=256$, GD , $lr=50$) — total pairs = 9,409

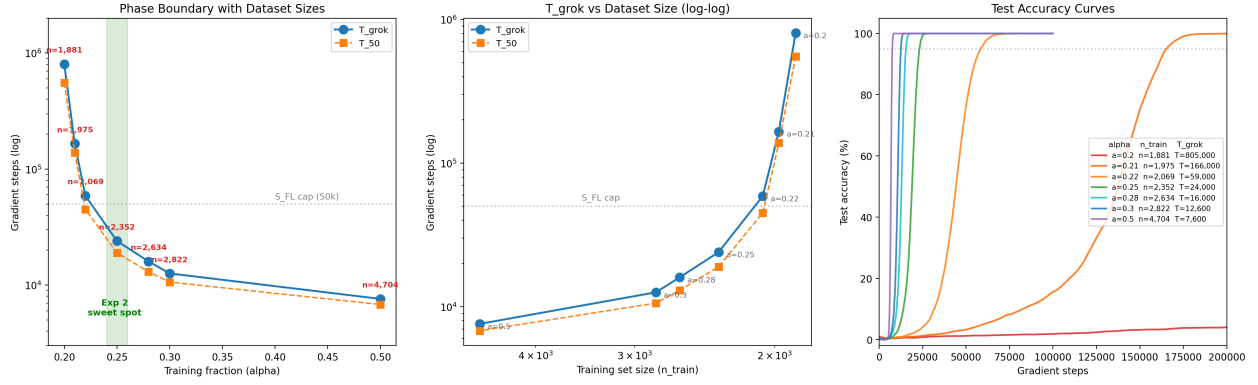


Figure 7: Refined phase boundary with dataset sizes (left), T_{grok} vs. training set size on log-log axes showing power-law scaling (centre), and test accuracy curves for all α values (right).

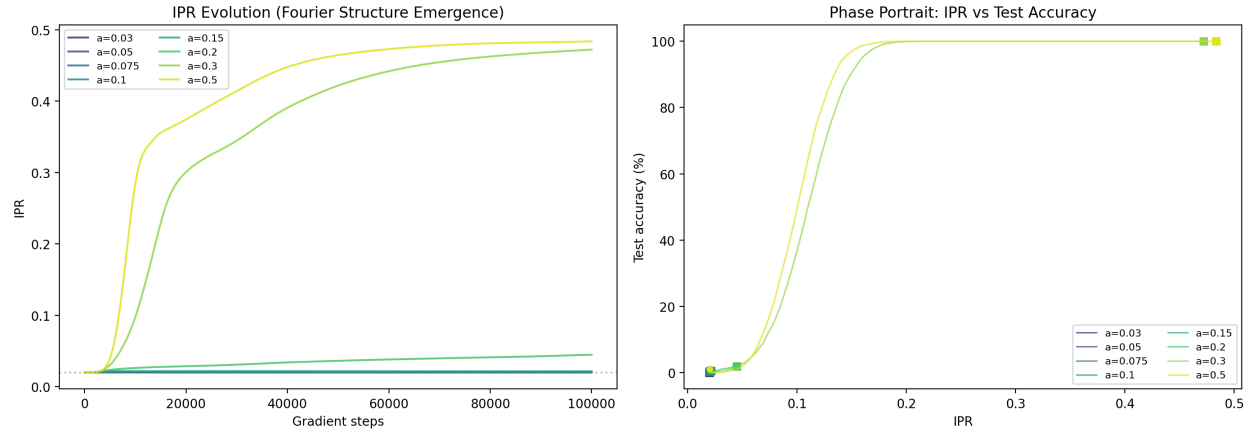


Figure 8: IPR evolution across α values (left) and phase portrait showing IPR vs. test accuracy (right). All α values that grok converge to the same attractor ($IPR \approx 0.35-0.48$).

B Centralised Phase Boundary: Extended Results

C Aggregation Effect: Additional Figures

D Grokking Anatomy Across Conditions

E Phase-Aligned Universal Fingerprint

F Drift as a Mechanistic Predictor

G Heterogeneity: Additional Results

H Optimisation Fragmentation (Experiment 4)

I FedProx Sweep

J Stress Tests: Client Scaling and Extreme Local Epochs

K Width Validation (Experiment 0)

L Seed Consistency

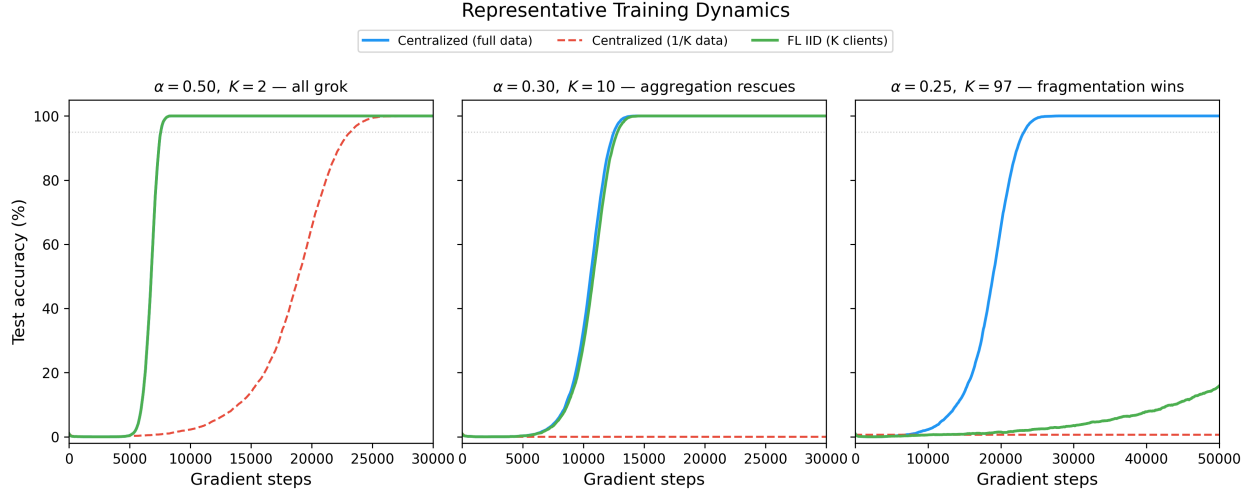


Figure 9: Test accuracy for three representative cells. Blue: centralised-full; red dashed: centralised-reduced; green: FL IID. In panel (b), 10 clients with individually hopeless data shards collectively recover grokking through aggregation alone.

Table 6: Client scaling ($\alpha=0.5$, IID, full participation). Grokking is immune to client count.

K	Samples/client	T_{grok}	Final acc	Final IPR
5	940	8,215	100.0%	0.342
10	470	8,220	100.0%	0.342
20	235	8,240	100.0%	0.340
50	94	8,280	100.0%	0.332
97	48	8,430	100.0%	0.311

Table 7: Extreme local epochs ($K=5$, IID). Even $E=200$ (100 rounds) groks. Higher E gives higher IPR.

Local epochs (E)	Rounds	T_{grok}	Final acc	Final IPR
50	400	8,300	100.0%	0.457
100	200	8,400	100.0%	0.446
200	100	9,000	100.0%	0.426

Table 8: Partial participation ($K=20$, target partition, $E=5$). The only axis that degrades accuracy.

Fraction (f)	Clients/round	T_{grok}	Final acc	Final IPR
1.0	20	9,585	98.5%	0.214
0.8	16	9,600	97.5%	0.218
0.6	12	9,725	96.2%	0.220
0.4	8	9,900	93.9%	0.222
0.2	4	9,720	94.5%	0.242

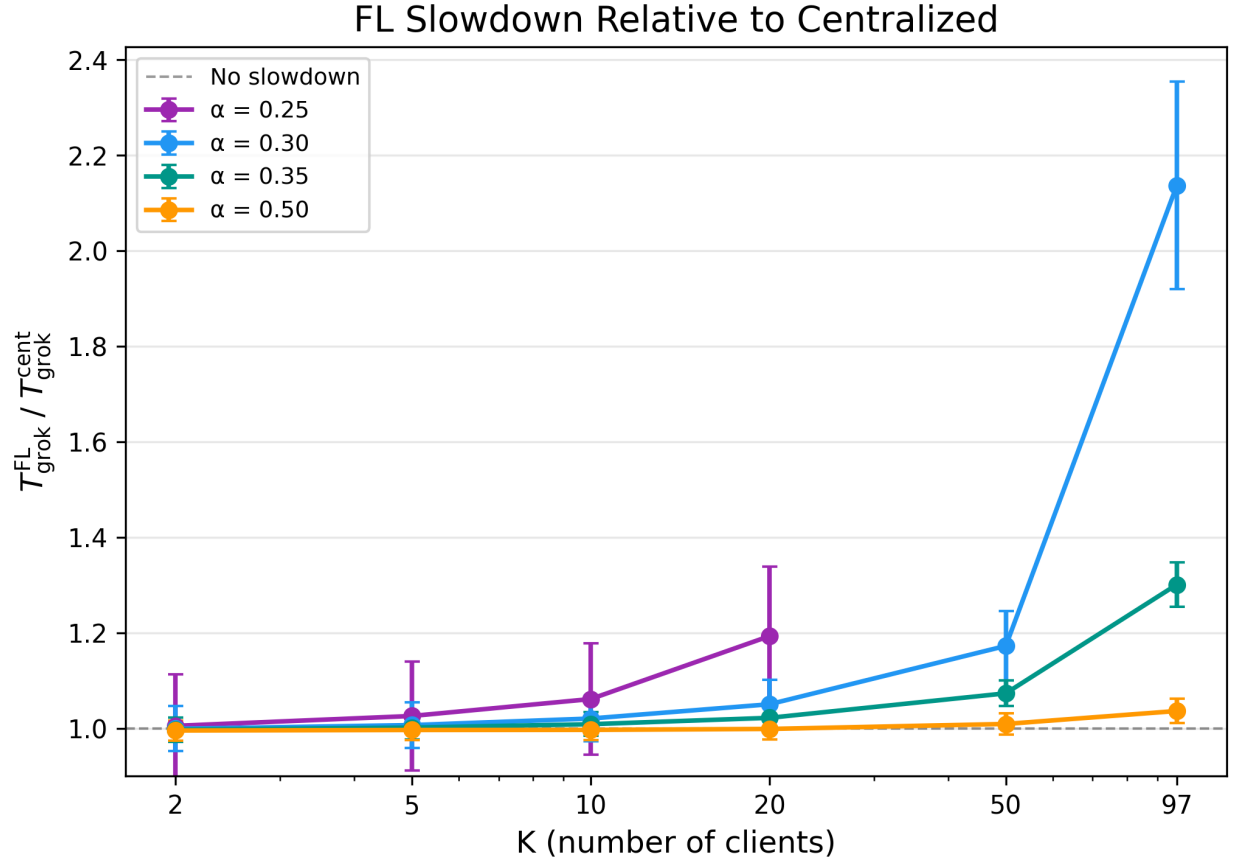


Figure 10: FL slowdown relative to centralised. Overhead is $< 1.2\times$ for $K \leq 20$ and only significant at $K = 97$ near the phase boundary ($\alpha = 0.25\text{--}0.30$).

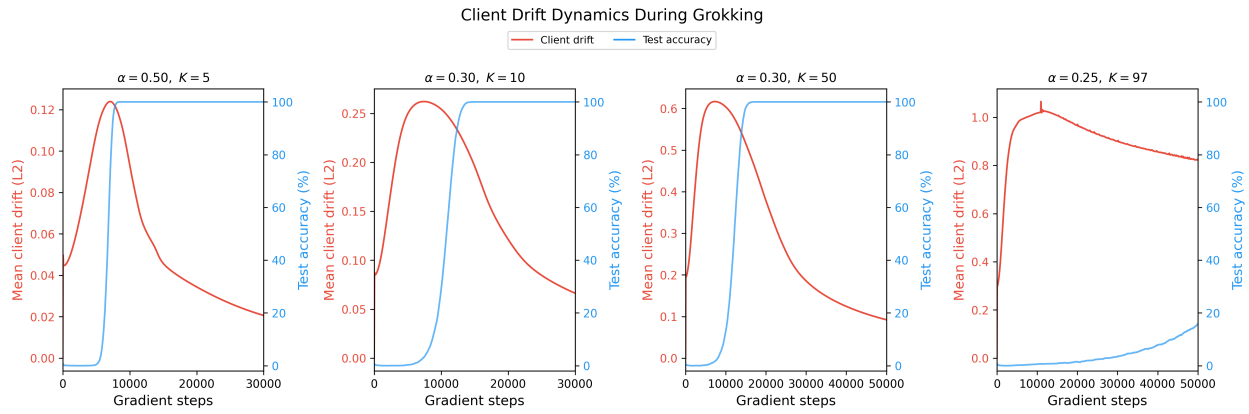


Figure 11: Client drift and test accuracy over time. Drift peaks at the grokking transition and drops once the Fourier solution locks in. At ($\alpha=0.25, K=97$), drift grows monotonically.

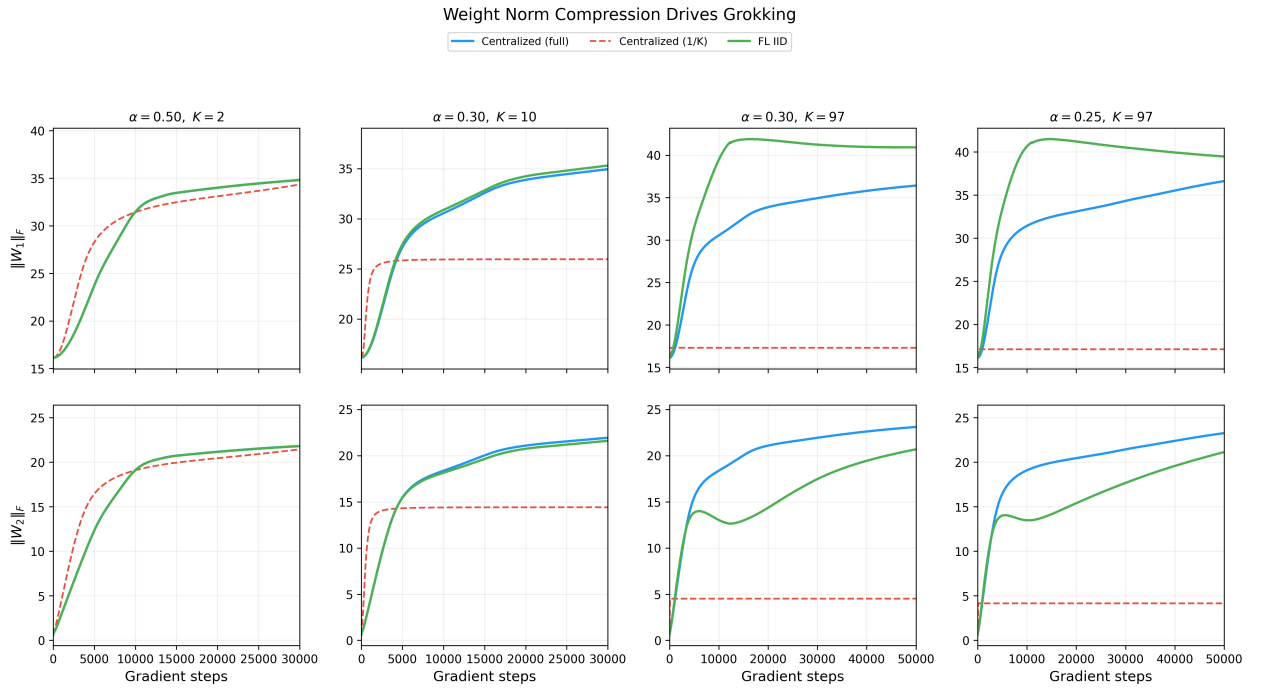


Figure 12: Weight norm evolution for layers 1 and 2 across conditions. Centralised-reduced (red dashed) plateaus at low norms—insufficient for grokking. FL IID (green) tracks centralised-full (blue), reaching the same magnitude.

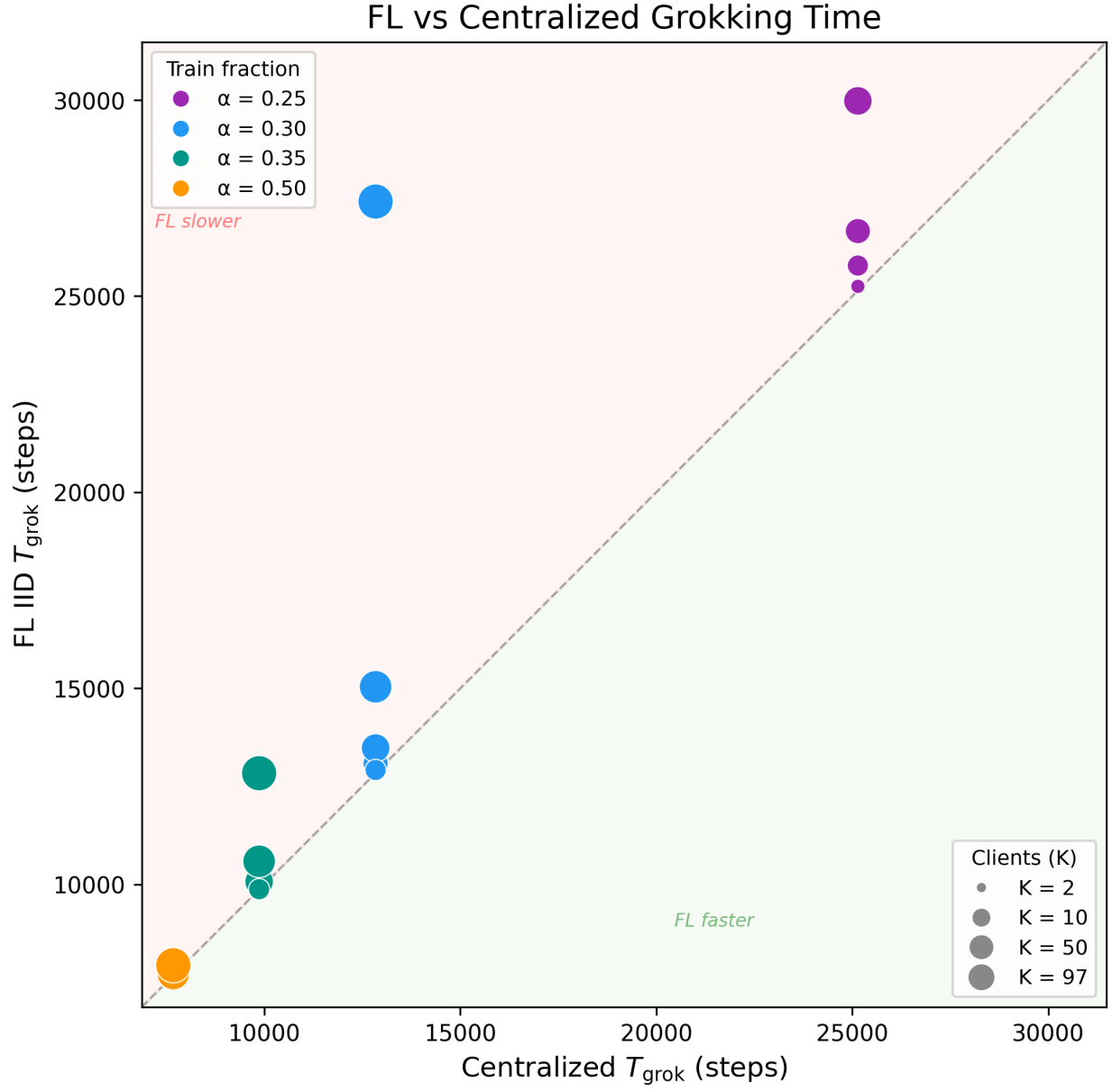


Figure 13: FL vs. centralised T_{grok} scatter. Points near the diagonal indicate minimal FL overhead; deviation grows with K and lower α .

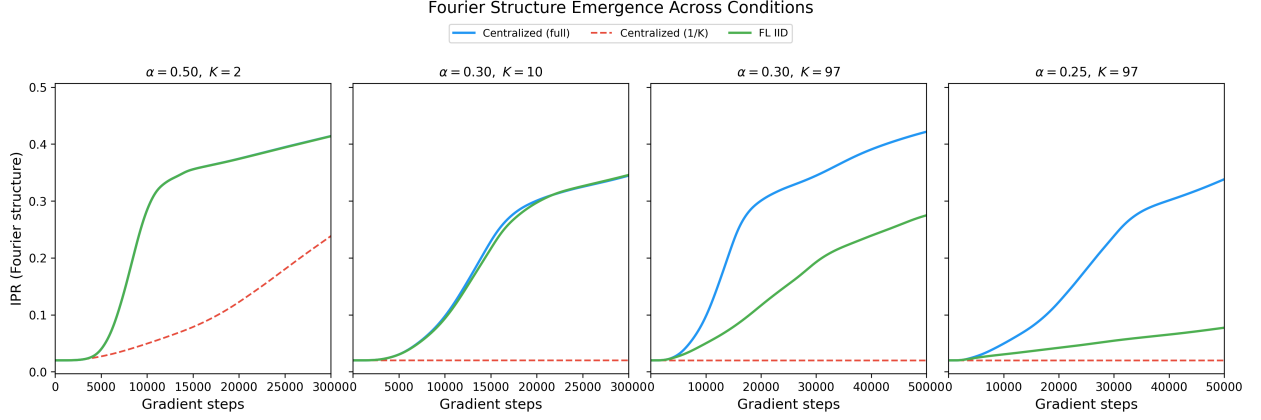


Figure 14: Fourier structure emergence (IPR) across conditions. FL IID tracks centralised-full; centralised-reduced fails to develop Fourier structure.

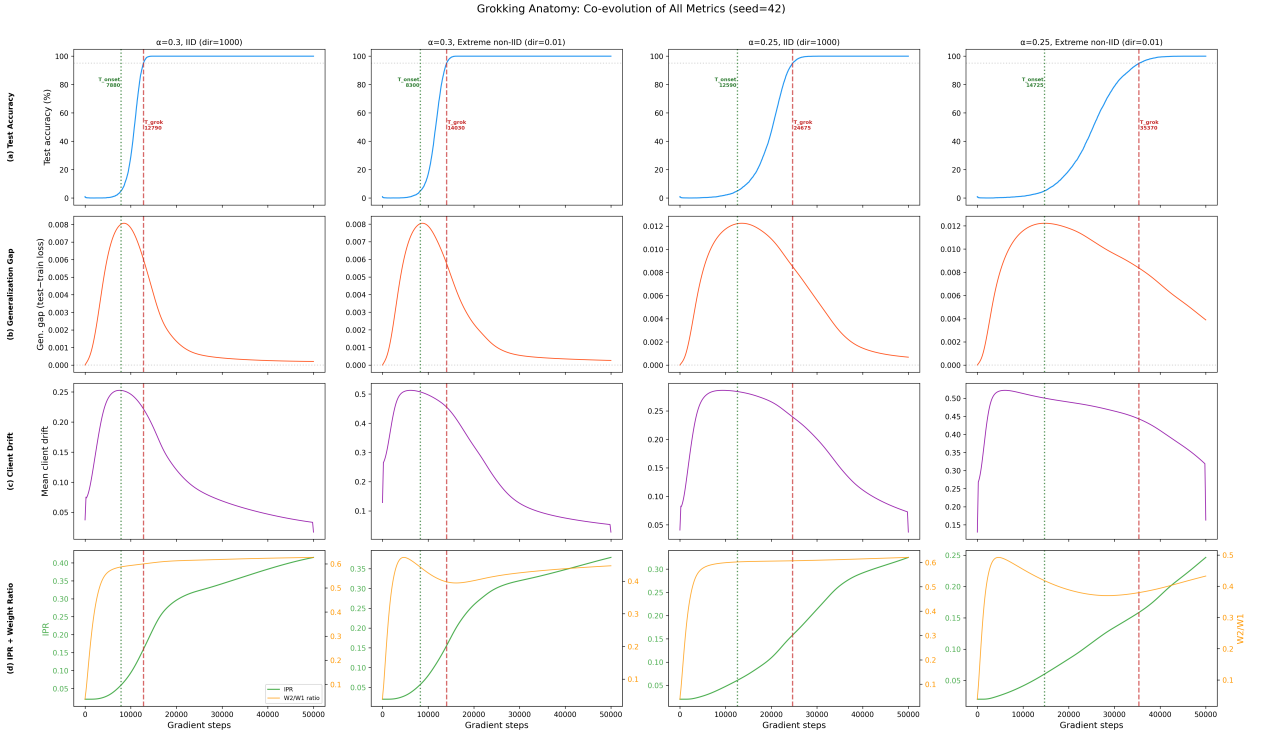


Figure 15: Grokking anatomy: four conditions from comfortable ($\alpha=0.3$, IID) to near-boundary ($\alpha=0.25$, $\alpha_{\text{dir}}=0.01$). Rows: test accuracy, generalisation gap, client drift, IPR, weight norms. Heterogeneity stretches the transition by up to 71% but the final outcome is identical.

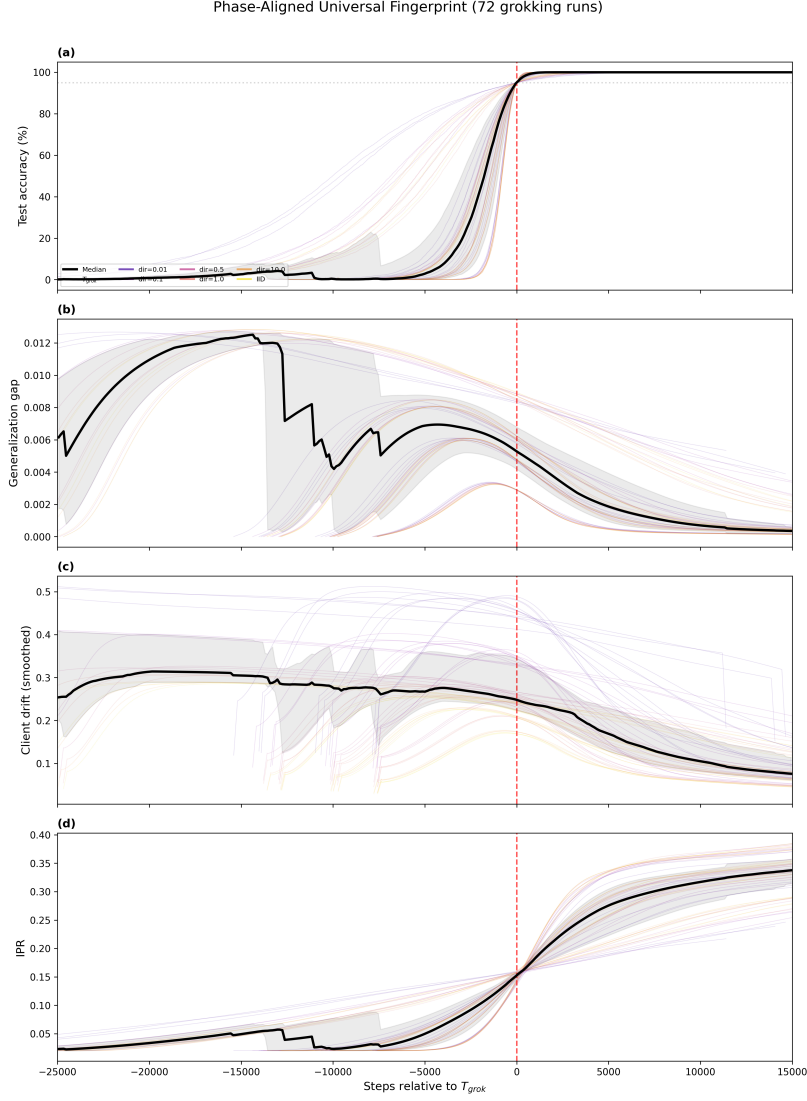


Figure 16: 72 grokking runs aligned to $T_{\text{grok}} = 0$. (a) Test accuracy, (b) generalisation gap, (c) client drift, (d) IPR. Black: median; shading: IQR. All runs follow a tight common trajectory despite spanning $K \in \{10, 20\}$, $\alpha_{\text{dir}} \in \{0.01-1000\}$, and $\alpha \in \{0.25-0.5\}$.

Figure 7: Drift Predicts Grokking Delay (Controlling for α)

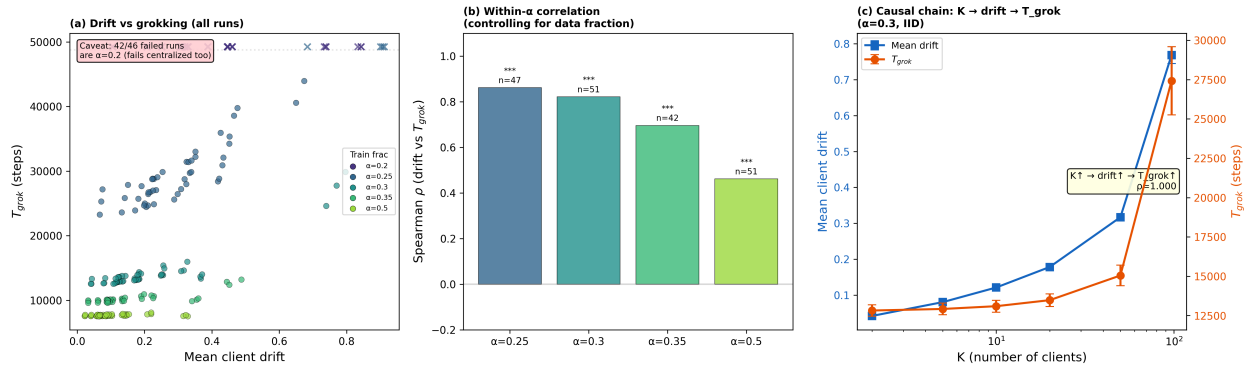


Figure 17: Drift predicts grokking delay, controlling for α . (a) T_{grok} vs. drift across all 237 runs (red box: 42/46 failures at $\alpha=0.2$). (b) Within- α Spearman ρ : significant at every α , strongest near the boundary. (c) Causal chain at $\alpha=0.3$: $K \rightarrow \text{drift} \rightarrow T_{\text{grok}}$.

Figure 6: Mechanistic Comparison of Representative Runs

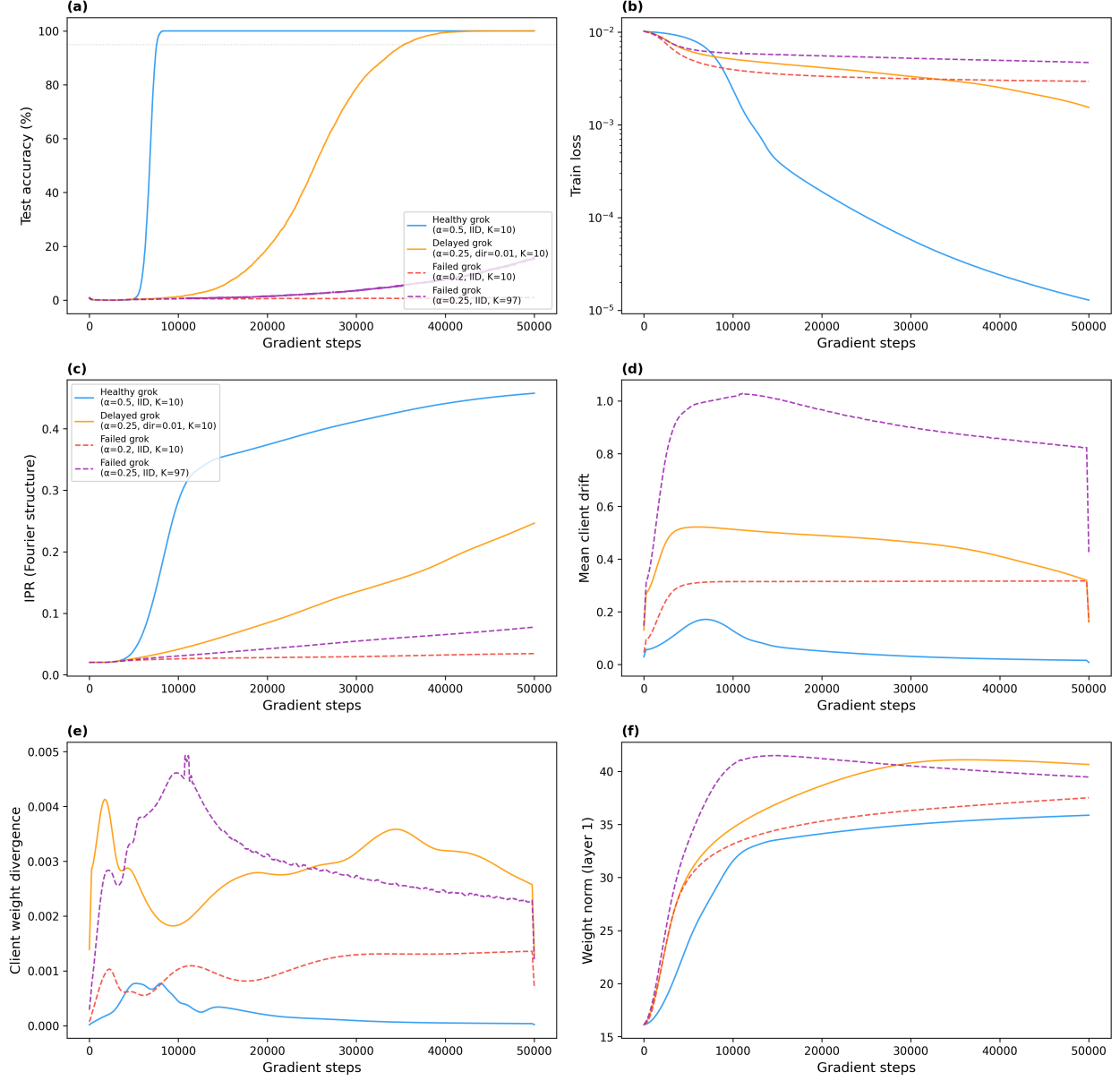


Figure 18: Representative run comparison. (a) Test accuracy, (b) train loss, (c) IPR, (d) mean client drift, (e) client weight divergence, (f) layer-1 weight norm. Healthy grok ($\alpha=0.5, K=10$), delayed grok ($\alpha=0.25, K=10$), and two failure modes (data-limited and fragmented).

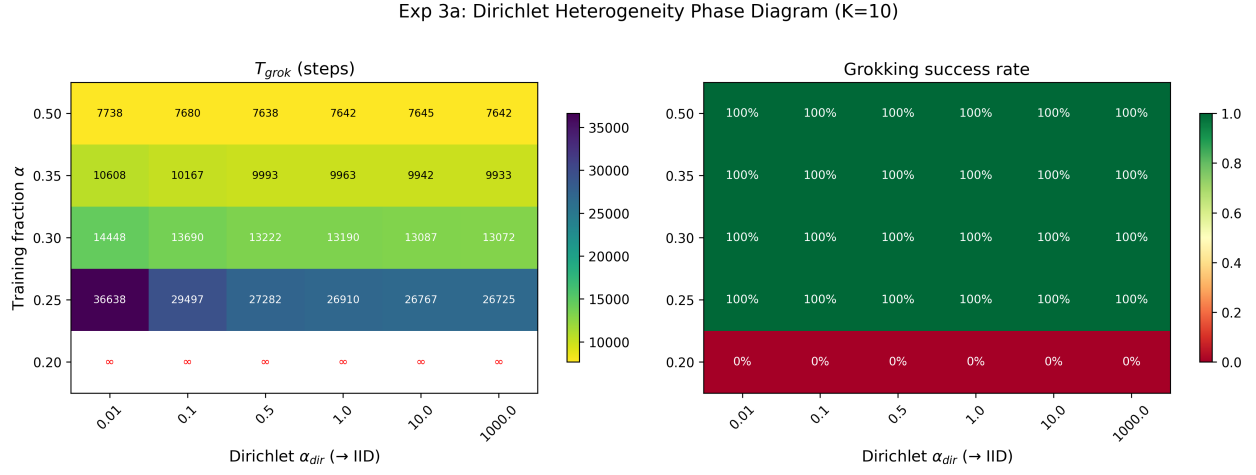


Figure 19: Exp 3a: Dirichlet heterogeneity phase diagram ($K=10$). Left: T_{grok} heatmap. Right: grokking success rate. The phase boundary is unchanged across all heterogeneity levels.

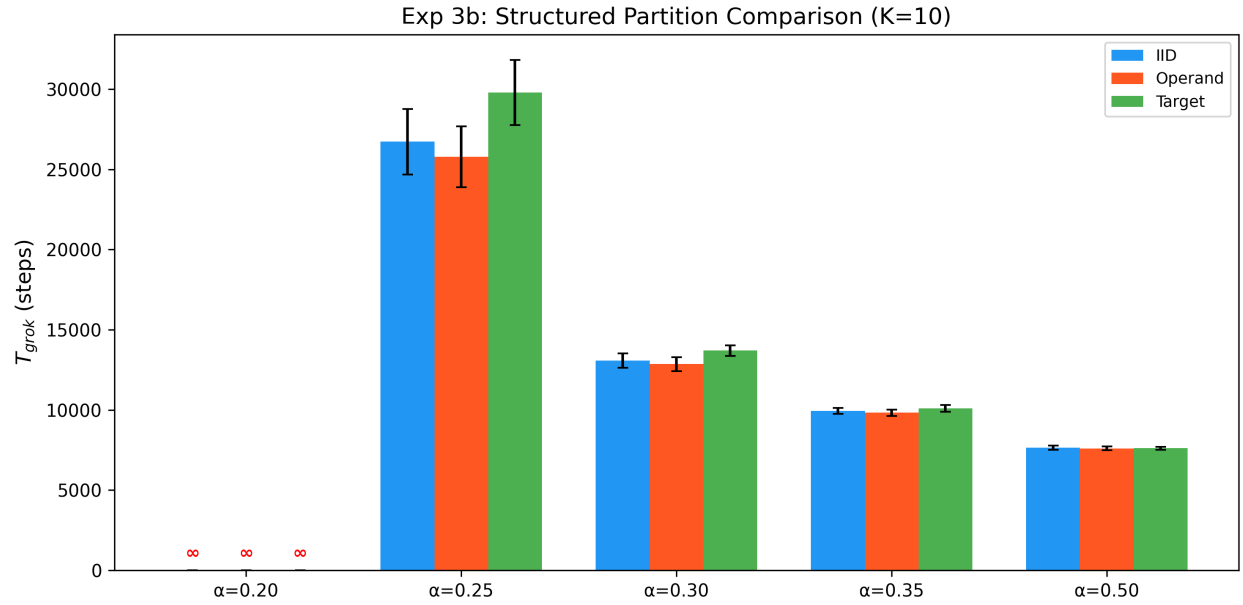


Figure 20: Exp 3b: Structured partition comparison ($K=10$). IID, operand-based, and target-based partitions yield statistically indistinguishable T_{grok} at all α values.

Weight Norm Evolution Across Heterogeneity Levels (seed=42)

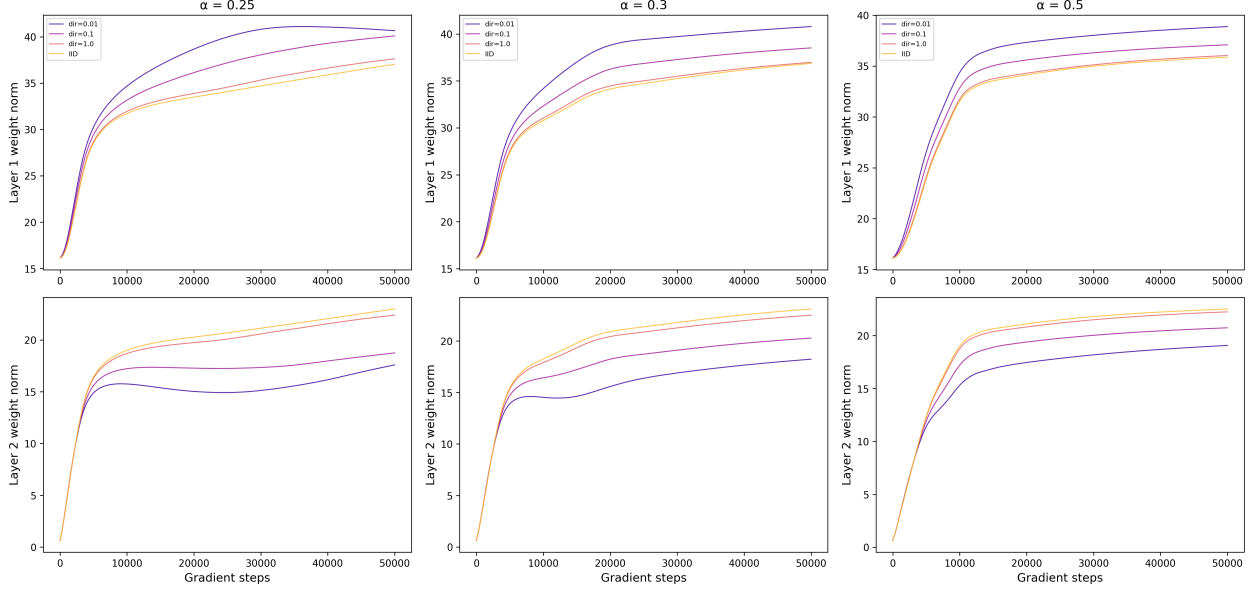


Figure 21: Weight norm evolution across heterogeneity levels. Non-IID slows growth by $\sim 20\text{--}30\%$ but final values converge.

IPR Trajectories by Heterogeneity Level (seed=42)

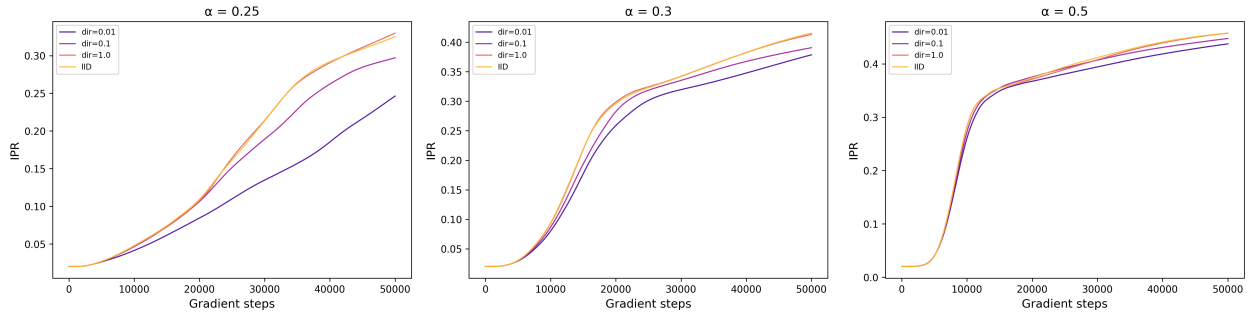


Figure 22: IPR trajectories by heterogeneity level. More heterogeneous data slows IPR growth but the final plateau is identical.

Exp 3a: Grokking Transition Sharpness (K=10)

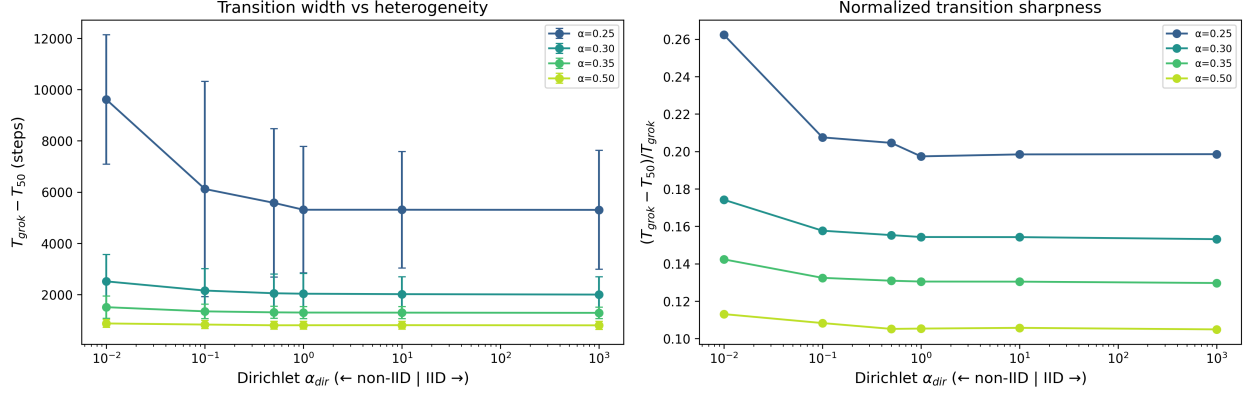


Figure 23: Transition sharpness vs. heterogeneity. Left: absolute width ($T_{\text{grok}} - T_{50}$). Right: normalised sharpness. Heterogeneity broadens the transition near the boundary but has minimal effect in the interior.

Drift-IPR Correlation at Grokking Transition (all seeds)

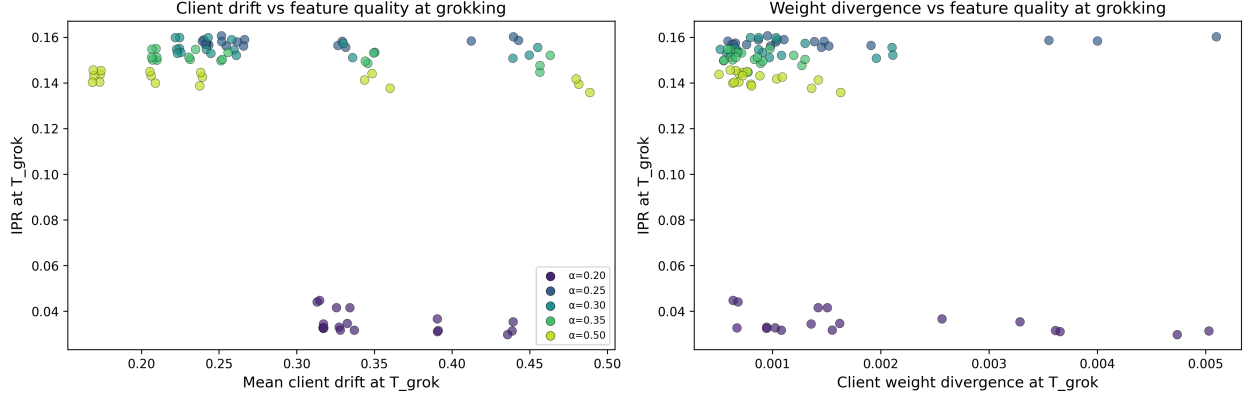


Figure 24: Drift-IPR correlation at the grokking transition. Left: client drift vs. IPR at T_{grok} . Right: weight divergence vs. IPR. IPR at grokking is invariant to drift level—the same Fourier solution is reached regardless of optimisation noise.

Exp 4a: Drift Accumulation $\times$ Heterogeneity (K=10, S=50k)

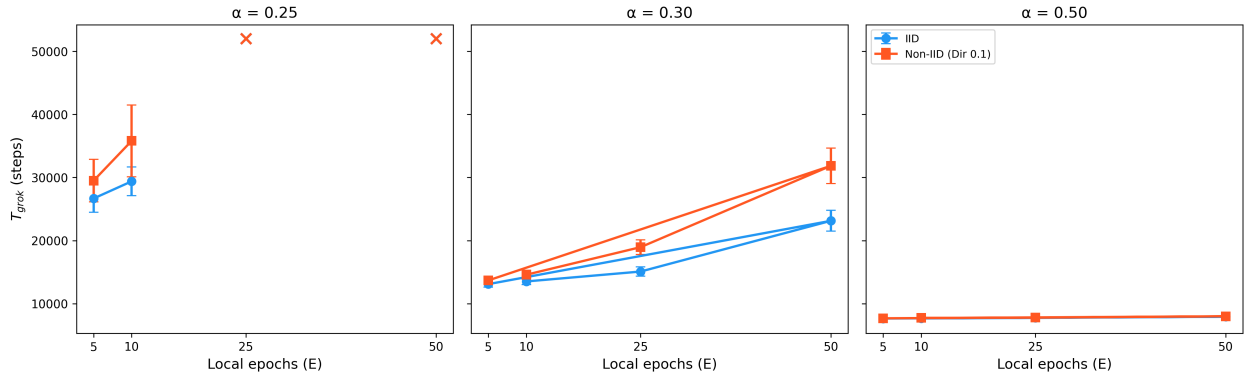


Figure 25: Exp 4a: Drift accumulation $\times$ heterogeneity ($K=10$, $S=50k$). At $\alpha=0.25$ with non-IID data, $E \geq 25$ prevents grokking (red $\times$). At $\alpha=0.5$, grokking is immune to local epochs.

Exp 4b: Partial Participation $\times$ Heterogeneity ($K=10, E=5$)

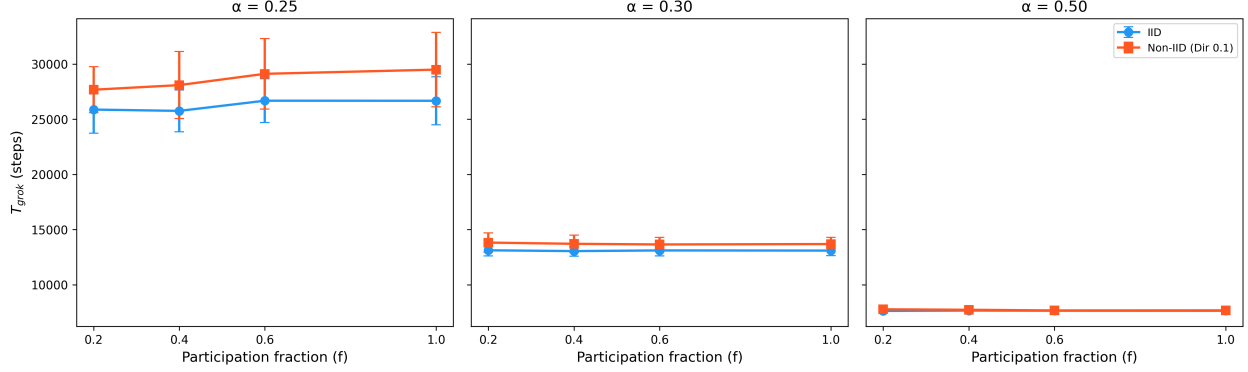


Figure 26: Exp 4b: Partial participation $\times$ heterogeneity ($K=10, E=5$). Participation fraction has minimal effect on T_{grok} at all α values, with slight delay under non-IID at $\alpha=0.25$.

Exp 4c: Compute vs Communication ($K=10, R=2000, \text{IID}$)

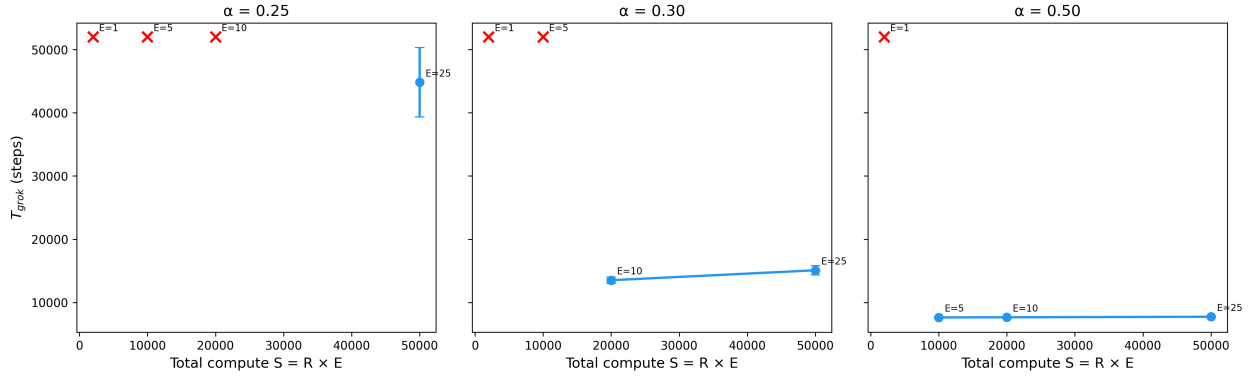


Figure 27: Exp 4c: Compute vs. communication tradeoff ($K=10, R=2000, \text{IID}$). At low α , insufficient local computation ($E=1$) prevents grokking (red $\times$). At $\alpha=0.5$, even $E=1$ suffices.

Experiment 0: Width Validation ($p=97, \text{GD}, \text{lr}=50, 50\text{k steps}$)

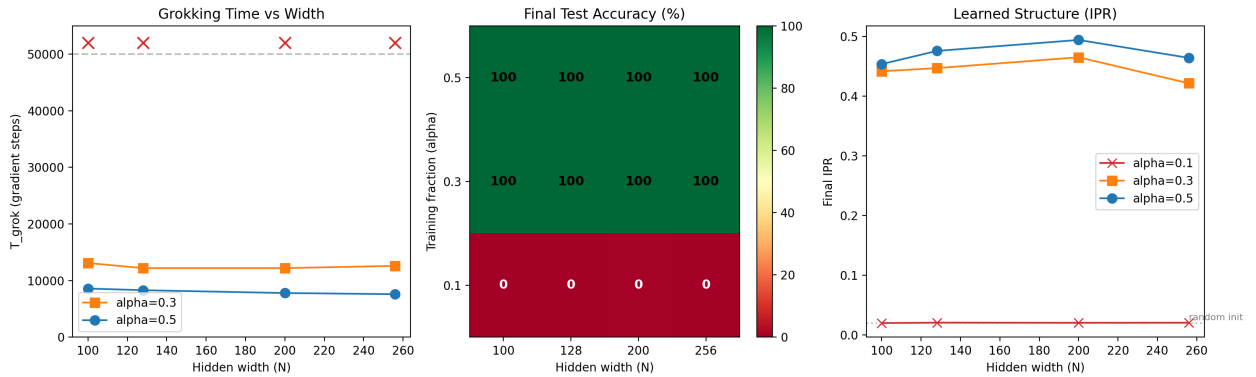


Figure 28: Width validation: $N \in \{100, 128, 200, 256\}$ across α values. All widths produce nearly identical T_{grok} ; width is not a confound.

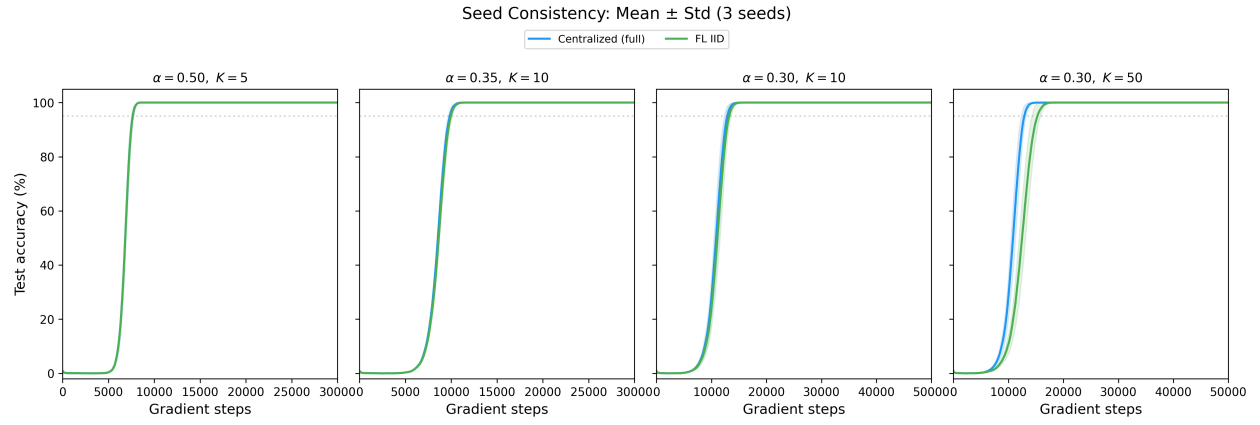


Figure 29: Exp 2: Seed consistency across the (α, K) grid. Low variance at high α ; increasing variability near the phase boundary.